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BANNARI AMMAN INSTITUTE OF TECHNOLOGY
(An Autonomous Institution Affiliated to Anna University, Chennai)
SATHYAMANGALAM - 638 401

Periodical Test - I - October 2025
I Semester

22MA101 & ENGINEERING MATHEMATICS I

Degree & Branch: B.E./B.Tech. & CS,AL

Time: 1:30 hrs

Maximum: 50 Marks

Instructions:

1. Students should not mark/write anything on the Question Paper other than the register number.
2. Section A and B of the Question Paper contains questions for 30 Marks each. Section C contains questions for 15 Marks.
3. Students can attempt answering any two out of three subsections in each section. Maximum mark is limited to 20 in section A & B, and 10 in sections C.

Q. No.	Questions				
Section - A	COURSE OUTCOME 1				MAXIMUM: 20 MARKS
A1	(i)	X: 1 2 3 4 5 Y: 2 3 5 4 6 The value of $\sum x^2$ is..... a) 10 b) 12 c) 15 d) 20 (1 Mark - [U/C, 1])			
	(ii)	Consider the following data: x: 1 2 3 4 5 y: 2 3 5 4 6 Find the value of $\sum x^2$ a) 40 b) 45 c) 50 d) 55 (1 Mark - [U/C, 2]) $1 + 4 + 9 + 16 + 25$			

(iii)

A person is trading in a share market daily. He is interested to find a relationship, as a mathematical equation, for his returns in a month with respect to the days of that month. The following data shows the profit (y) with respect to the days (x). The amount is given in terms of thousands of Rupees. For a straight line of the form $y = ax + b$, identify the values of a and b.

x: -4

1

2

3

y: 4

6

8

10

Hint: From the method of least squares, the normal equations are

$$a \sum x + nb = \sum y; a \sum x^2 + n \sum x = \sum xy$$

(4 Marks - [Ap/P, 2])

(iv)

A person is trading in a share market daily. He is interested to find a relationship, as a mathematical equation, for his returns in a month with respect to the days of that month. The following data shows the profit (y) with respect to the days (x). The amount is given in terms of thousands of Rupees. Find the mathematical model as a straight line of the form $y = ax + b$.

x: 0

1

2

3

4

y: 1

1.8

3.3

4.5

6.3

Hint: From the method of least squares, the normal equations are

$$a \sum x + nb = \sum y; a \sum x^2 + n \sum x = \sum xy$$

(4 Marks - [Ap/P, 2])

A2

(i)

Identify the determinant of the matrix A, if the characteristic equation of the matrix A is

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6$$

a) 6

b) -6

c) 11

d) -11

(1 Mark - [R/F, 1])

(ii)

The sum of the Eigen values of the matrix

 A^2

is _____ if

$$A = \begin{pmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$$

a) 10

b) 100

c) 38

d) 36

(1 Mark - [U/C, 2])

(iii)

Consider two currents

 I_1

= [4, 1] A and

 I_2

= [-2, 2] A in a circuit. If the total current is a linear combination of these two currents, given by

$$I = aI_1 + bI_2$$

, find the constants a and b if the total current is $I = [2, 8]$ A.

(iv)

Consider a transportation network consisting of three cities: City A, City B, and City C. The matrix A represents the transportation connectivity matrix, where each element $A[i][j]$ represents the transportation flow or traffic volume between City i and City j. Here's an example matrix: In this transportation network analysis scenario, the matrix A represents the flow of goods, passengers, or vehicles between cities.

$$M = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

Similarly, the other elements of the matrix A represent the transportation flows between different cities. The characteristic equation for the above transportation network matrix is

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

The Eigen values are: 1, 2, 3 and the corresponding Eigen vectors are:

$$\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

The largest Eigen value is 3 and the corresponding Eigenvector is

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

here 1 is the highest value in this Eigen vector and the corresponding cities are A and C. This indicates that the transportation demand or flow is highest among the most significant Eigen vector. In this case, City A and C have the highest flow value, followed by City B. Cities A and C are likely a central hub or receives the highest traffic volume in the transportation network. Based on the above scenario, consider the transportation network matrix A which represents the flow of goods, passengers, or vehicles between cities:

$$\begin{pmatrix} 0 & -2 & -2 \\ -1 & 1 & 2 \\ -1 & -1 & 2 \end{pmatrix}$$

Find the transportation hub using Eigen values and Eigen vectors.
Hint: The characteristic equations is

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

(5 Marks - [Ap/P, 2])

A3

(i)

x: 1 2 3 4 5
y: 60 65 70 75 80

The linear model of the form $y = 5x + 55$, the value of y at $x = -1$ is

- a) 50
- b) 51
- c) 53
- d) 55

(1 Mark - [U/C, 2])

(ii)

Find the average value of Y using the given data.

X 2 4 6 8

	Y	10	20	30	40
a) 10					
b) 22					
c) 25					
d) 29					

(1 Mark - [U/P, 1])

- (iii) A financial analyst wants to analyze the relationship between the age of a car (X) and its resale value (Y). Understanding this relationship can help him to estimate the depreciation rate of cars over time and make informed decisions regarding car purchases and sales. To investigate this relationship, you collect data on the ages and corresponding resale values of several cars. Here are the data points you obtained:

Age of cars (x) in years	2	3	4
Resale value(y) in Thousands	9	7	5

Compute the regression equation of age of cars on resale value given that $b_{xy} = -0.5$.
HINT:

$$x - \bar{x} = b_{xy} (y - \bar{y})$$

(3 Marks - [Ap/P, 2])

- (iv) Ravi working for an agricultural company. He wants to predict the crop yield (in tons per hectare) based on the amount of water used (in litres per hectare). He collected the following data from several fields:

Water used (liters/ha) (x)	100	200	300	400
Crop Yield (tons/ha) (y)	2.5	3.5	4.5	5.5

Compute the regression equation and predict the crop yield when 180 liter/ha and 340 liter/ha of water is used given that $b_{yx} = 0.01$.
HINT:

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

(5 Marks - [Ap/P, 2])

Section - B

COURSE OUTCOME 2

MAXIMUM: 20 MARKS

B1

(i)

Consider the following data

x	30	40	50	60	70
y	6.5	6	6.2	7.8	10.6

The value of

$$\sum xy$$

is

- a) 1955
b) 1965
c) 1975
d) 1985

(1 Mark - [U/C, 1])

(ii)

The parabola represented by the quadratic function

$$f(x) = ax^2 + bx + c$$

with 'a > 0' opens _____

- a) Upward
- b) Downward
- c) Leftward
- d) Rightward

(1 Mark - [U/C, 1])

(iii)

In stock market analysis, x could represent risk (variance or volatility), and y could represent returns. A negative value for x could indicate negative volatility (unfavorable fluctuations), while a negative y represents negative returns (losses).

x	-3	-2	-1	0	1	2	3
y	-5	-1	0	1	3	4	4

The table of observations is given above. From the above table, we have

$$\Sigma x = 0, \Sigma x^2 = 28, \Sigma x^3 = 0, \Sigma x^4 = 196, \Sigma y = 6, \Sigma xy = 40, \Sigma x^2 y = 6$$

Find the polynomial

$$y = ax^2 + bx + c$$

which represents the growth of a plant, using quadratic polynomial of best fit.

Hint: The normal equations are

$$\Sigma y = a \Sigma x^2 + b \Sigma x + nc;$$

$$\Sigma xy = a \Sigma x^3 + b \Sigma x^2 + c \Sigma x;$$

$$\Sigma x^2 y = a \Sigma x^4 + b \Sigma x^3 + c \Sigma x^2$$

(3 Marks - [U/C, 2])

(iv)

A group of students start a small company that produces CDs. The weekly profit (in dollars) is given by the function

$$p(x) = -2x^2 + 80x - 600$$

where x represents the number of CDs produced.

- a) Find the break-even points (x-intercepts) and y-intercept of the profit function.
- b) Find the vertex of the profit function, and interpret its meaning in the context of this problem.
- c) Sketch the profit function.

(5 Marks - [U/C, 2])

B2

(i)

The quadratic form $Q(x,y) = 2x^2 - 6xy + 3y^2$ corresponds to the matrix is

a)

$$\begin{bmatrix} 2 & -6 \\ 0 & 3 \end{bmatrix}$$

b)

$$\begin{bmatrix} 2 & -3 \\ -3 & 3 \end{bmatrix}$$

	d)	$\begin{bmatrix} 2 & -1.5 \\ -1.5 & 3 \end{bmatrix}$ $\begin{bmatrix} 2 & -2 \\ -2 & 3 \end{bmatrix}$ (1 Mark - [U/C, 1])
	(ii)	In a control systems problem, the quadratic form representing the system's energy has the canonical form $3x_1^2 - 4x_2^2 + x_3^2$ The index of system's energy is a) 0 b) 1 ☒ c) 2 d) 3 (1 Mark - [R/F, 1])
	(iii)	The stability of a drone flight control system is being analyzed. The quadratic form representing the stability matrix $A = \begin{pmatrix} 3 & 2 & 1 \\ 0 & -1 & 4 \\ 0 & 0 & 2 \end{pmatrix}$ a) Compute the eigenvalues of the system's stability matrix b) Construct the canonical form of the stability matrix using its eigenvalues. c) Find the rank and nature of the system's stability based on the eigenvalues and canonical form (4 Marks - [U/C, 1])
	(iv)	An economist is modelling an optimization problem where the profit function is represented by a quadratic form $2xy + 2yz + 2xz.$ The eigenvalues of the matrix representing the quadratic profit function are crucial for understanding the firm's profit behaviour: Positive eigenvalues: Stable and optimal profit point. Negative eigenvalues: Decreasing profits with increased input. Mixed (indefinite): Complex situation with trade-offs and the potential for saddle points, where the firm must carefully balance inputs. a) Convert the quadratic form into matrix form. b) Find the eigenvalues of the matrix and interpret how they relate to the nature of the motion. (4 Marks - [Ap/P, 2])
B3	(i)	Which among the following is not a type of quadratic form? a) Positive definite b) Negative definite c) Indefinite ☒ d) Partial definite (1 Mark - [R/F, 1])
	(ii)	If A is an orthogonal matrix then = A^{-1} $AA^T = I$ $A = \frac{I}{A^T}$ $A^{-1} = A^T$

	a) A^T b) A c) 1 d) $(A^{-1})^{-1}$ (1 Mark - [R/F, 1])
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(iii)	In a robotics problem, the motion of a robot is modelled using a quadratic form $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz.$ Finding the eigenvalues allows us to analyze the stability and nature of the robot's motion Positive eigenvalues: the robot exhibits stable motion. Negative eigenvalues: the robot's motion is stable, but rather than oscillating or continuing indefinitely, the robot's motion will decay over time Mixed (indefinite): the motion is unstable in at least one direction. a) Covert the quadratic form into matrix form. b) Find the eigenvalues of the matrix and interpret how they relate to the nature of the motion. (4 Marks - [Ap/P, 2])
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(iv)	A team of researchers is studying the spread of information within a social network, and they use matrix A to model the influence between two key individuals. Each person represents a node in the network, and the interactions between them are captured in the matrix. The rows and columns of A correspond to Person 1, Person 2 respectively. The matrix is given by $A = \begin{pmatrix} -2 & 2 \\ 2 & 1 \end{pmatrix}$ a) If the eigen value is -3, the corresponding eigen vector is $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$ Find the Eigen vectors (it will indicate which individuals are most influential or most affected by the network's collective dynamics) corresponding to the eigen value 2. Hint: $\begin{pmatrix} -2 - \lambda & 2 \\ 2 & 1 - \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ b) Diagonalize the matrix A to reveal the independent modes of influence in the network. Hint: $N^T A N = D$ (4 Marks - [Ap/P, 2])
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Section - C		COURSE OUTCOME 3	MAXIMUM: 10 MARKS
C1	(i)	For a cubic polynomial $y = ax^3 + bx^2 + cx + d$, how many coefficients are fitted? a) 2 b) 3 c) 4 d) 5 (1 Mark - [U/C, 1])	
	(ii)	The height H of a plant (in cm) at different times t (in days) is recorded. Assume that the relationship between the height H and the time t follows a power curve $H = a t^b$	

The data is given below:

Time (t) (days)	Height (H) (cm)
2	5
4	9
6	15
8	24
10	35

Frame the normal equations related to the plant's height H and time t.

HINT: The logarithmic transformation is

$$\log H = \log a + b \log t$$

$$\text{Let } Y = \log H ; A = \log a \text{ and } X = \log t$$

The normal equations are

$$\sum Y = nA + b \sum X$$

$$\sum XY = A \sum X + b \sum X^2$$

(4 Marks - [U/P, 2])

C2

(i)

If after applying least squares, the power curve is found to be

$$2.5X^{10}$$

, What is the value of a and b ?

a) a=2.5, b=5

b) a=8.1, b=15

☒ c) a=2.5, b=10

d) a=25, b=5

(1 Mark - [U/C, 1])

(ii)

A speeding motorist slams on the brakes when she sees an accident directly ahead of her. The distance she has travelled t seconds after braking is shown in the table. Fit the Normal equations of the third degree polynomial.

Time seconds (x)	0	1	2	3	4
Distance in feet (y)	1	1.8	1.3	2.3	6.3

$$\sum y = na + b \sum x + c \sum x^2 + d \sum x^3$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3 + d \sum x^4$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4 + d \sum x^5$$

$$\sum x^3 y = a \sum x^3 + b \sum x^4 + c \sum x^5 + d \sum x^6$$

(4 Marks - [U/P, 2])

C3

(i)

If after applying least squares, the power curve is found to be

$$y = 5x^{0.75}$$

what does this indicate about the relationship between y and x?

- a) y increases linearly with x
- b) y decreases as x increases
- c) y increases more slowly than x
- d) y is directly proportional to x

(1 Mark - [U/C, 1])

(ii)

A computer scientist is analyzing the relationship between processor speed S (in GHz) and performance P (in millions of instructions per second, MIPS) of a system

$$P = aS^b$$

The data is:

Speed (GHz)	1	2	3	4	5
Performance (MIPS)	100	210	360	590	820
$\log_{10}(S)$	0.000	0.301	0.477	0.602	0.699
$\log_{10}(P)$	2.000	2.322	2.556	2.771	2.914

If the normal equations are $5A + 2.079b = 12.563$; $2.079A + 1.169b = 5.623$
Find the value of a and b and fix the power curve.

(4 Marks - [U/P, 2])

*** End of Question Paper ***